

Assumptions, dummies, logs and curves

International Marketing Analytics

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Today

Assump-tions | Dummy coding | Log-log | Non-linear | Python | In class

Theory, Parts 1 to 4

- Six assumptions: look, test, fix
- Dummies for categories, the reference group
- Log-log and the other log forms
- Curves: squares, logs, the turning point

Practice, Parts 5 and 6

- statsmodels tests, polars dummies, plotnine plots
- Online quiz 3 and its answers
- Coding exercise in pairs, then the answers
- The same tasks with Posit Assistant

Readings: Skiera, Reiner & Albers (2022), Regression Analysis; Bojinov, Parzen & Hamilton (2025), Linear Regression (HBS).

Where we are: two models for brand 1

```
linear:  sales    = b0 + b1·price1 + b2·feature1 + b3·display1 + b4·temp + b5·december
log-log: ln sales = b0 + b1·ln price1 + b2·ln price2 + b3·ln price3 + b4·ln price4
                  + b5·feature1 + b6·display1 + b7·temp + b8·december
```

	Linear	Log-log
Own price	−678 units per EUR	−3.68 (elasticity)
R^2	0.896	0.903
Outcome	sales (units)	log sales

Same data, two models: which one may we trust? R^2 cannot tell, because the outcomes differ. The residuals can.

Sources: make_assumptions_figures.py on data/legacy/chocolate_dataset.csv; deck 01b, Parts 6 to 9

Assumptions and how to test them

Part 1 of 6 · look first, then test, then fix

Six assumptions, three steps each

Assumption	Look	Test (H0: assumption holds)	Fix
1 Linear form	residuals vs fitted: curve?	RESET	logs, squares
2 Constant variance	residuals vs fitted: funnel?	Breusch-Pagan	logs; HC3 errors
3 Normal residuals	Q-Q plot, histogram	Jarque-Bera, Shapiro-Wilk	logs; large n helps
4 Independent errors	residuals over time	Durbin-Watson ≈ 2	HAC errors; Session 4
5 No multicollinearity	correlation matrix	VIF below 5 (10)	drop, combine, more data
6 No dominant week	Cook's distance by week	above 4/n: inspect	check data, with and without

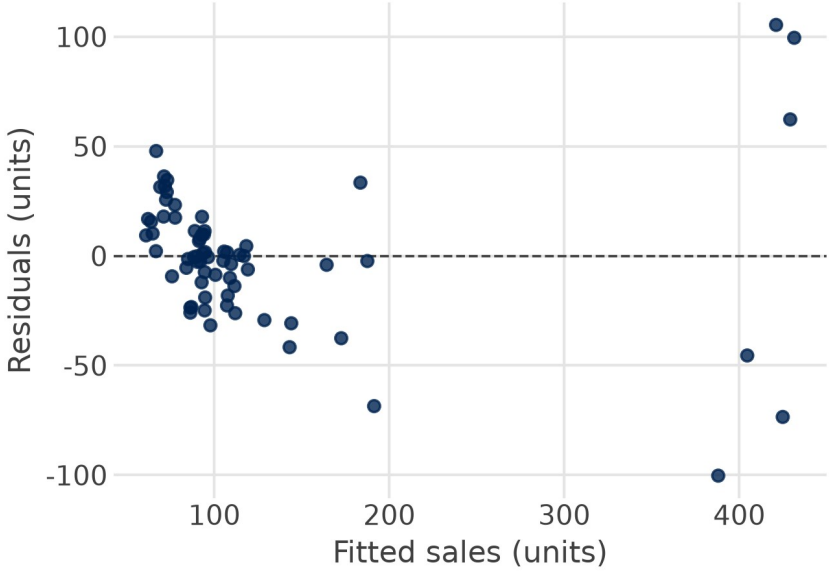
Sources: Skiera, Reiner & Albers 2022; Wooldridge 2025, ch. 8 to 9; Hair et al. 2019

How to read an assumption test

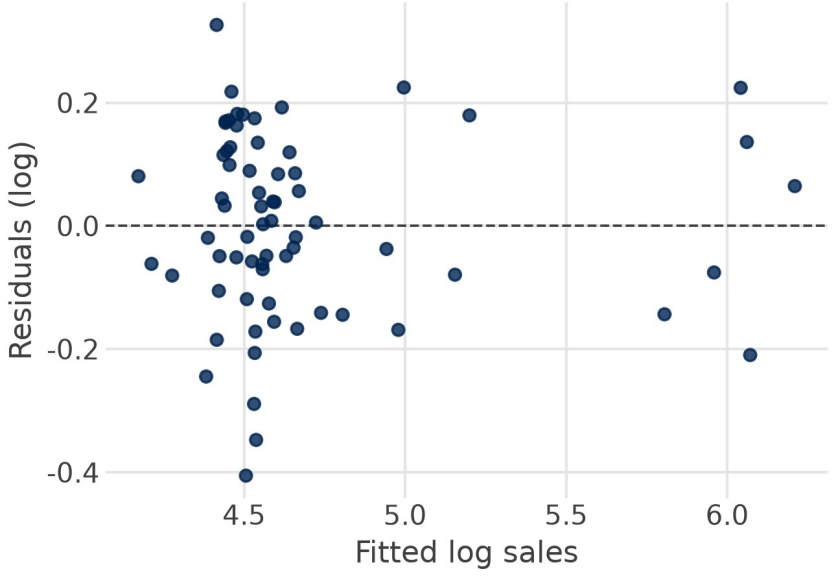
1. **H0 = the assumption holds:** constant variance, normal residuals, the right form
2. **p below 0.05:** evidence against H0, so the assumption is violated: act
3. **p above 0.05:** no evidence of a problem; not a proof that all is well
4. **Always look as well:** small samples miss problems, very large samples flag tiny ones

⚠ **Mind the direction:** in a coefficient test a small p is good news (an effect); in an assumption test it is bad news.

1 Linear form: look for a curve



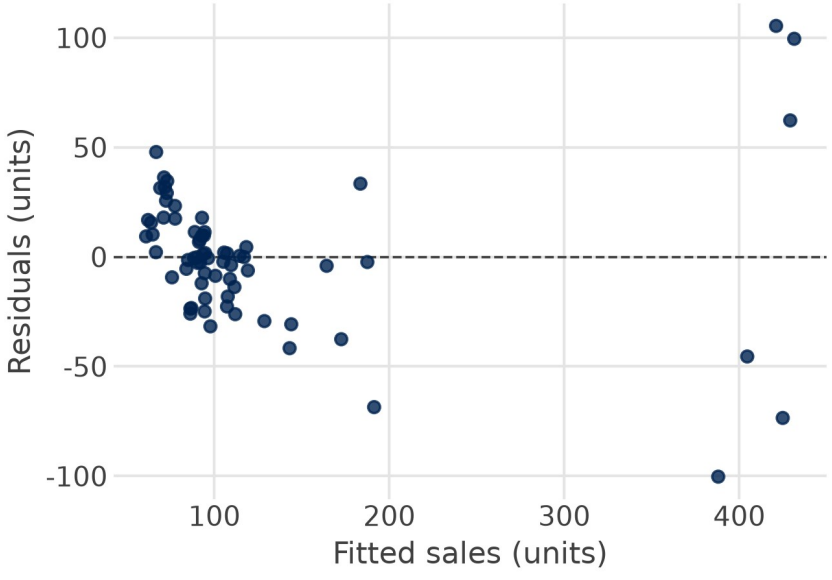
Linear model: a U-shape. RESET p < 0.001.



Log-log model: no curve. RESET p = 0.32.

RESET: adds the squared fitted values; if they help, the form is wrong.

2 Constant variance: look for a funnel



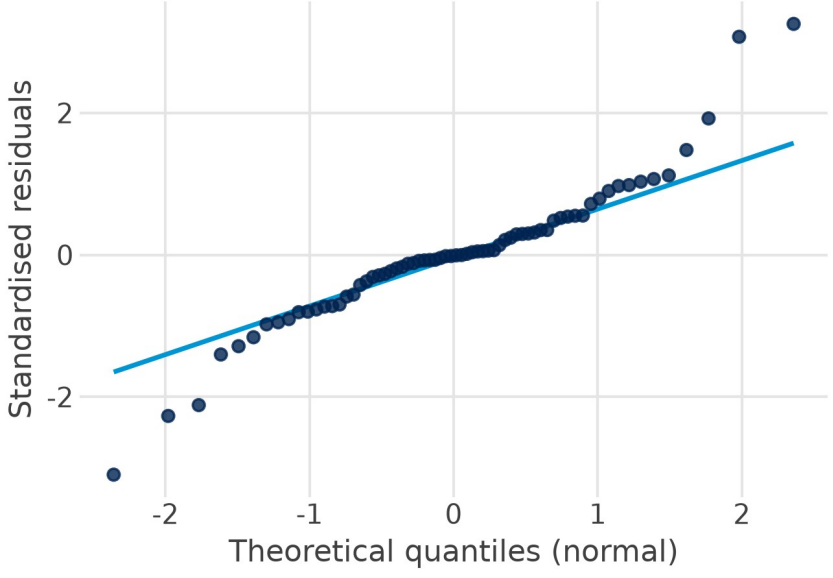
< 0.001 Breusch-Pagan p, linear model: unequal variance

0.68 Breusch-Pagan p, log-log model: no evidence

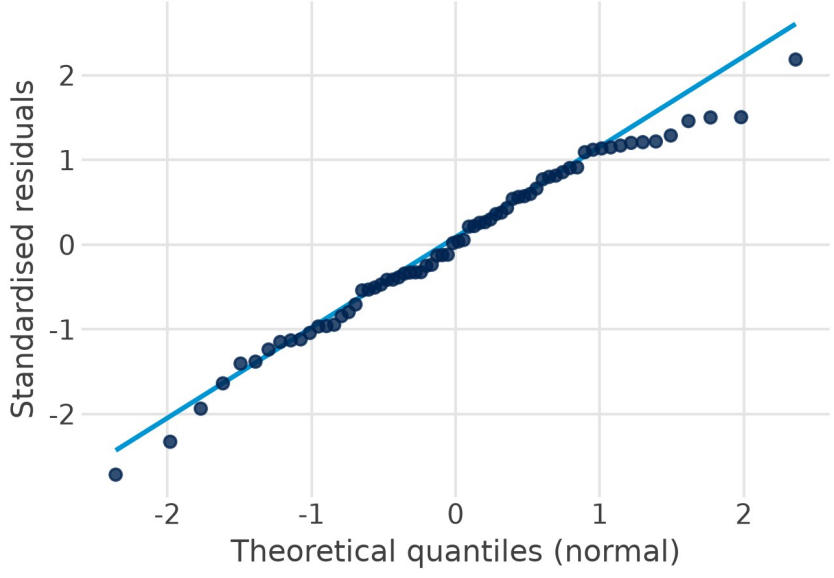
110 se of price1 in the linear model with HC3 errors (usual: 44)

Fix: logs, because sales vary in per cent; or robust errors with fit(cov_type="HC3").

3 Normal residuals: the Q-Q plot



Linear: heavy tails. Jarque-Bera $p < 0.001$.



Log-log: on the line. Jarque-Bera $p = 0.50$.

Matters least: with many observations the tests still work; extreme weeks matter more.

4 Independent errors: weeks follow weeks

≈ 2	Durbin-Watson without autocorrelation (range 0 to 4)
1.13	log-log model: positive autocorrelation between weeks
0.31	se of the elasticity with HAC errors, 1 lag (usual: 0.27)

Theory only today: Session 4 (time series) models autocorrelation with lags. Not in the exercise or the homework.

5 Multicollinearity: the VIF

$$VIF_j = 1 / (1 - R^2_j)$$

R^2_j: variable j regressed on all others

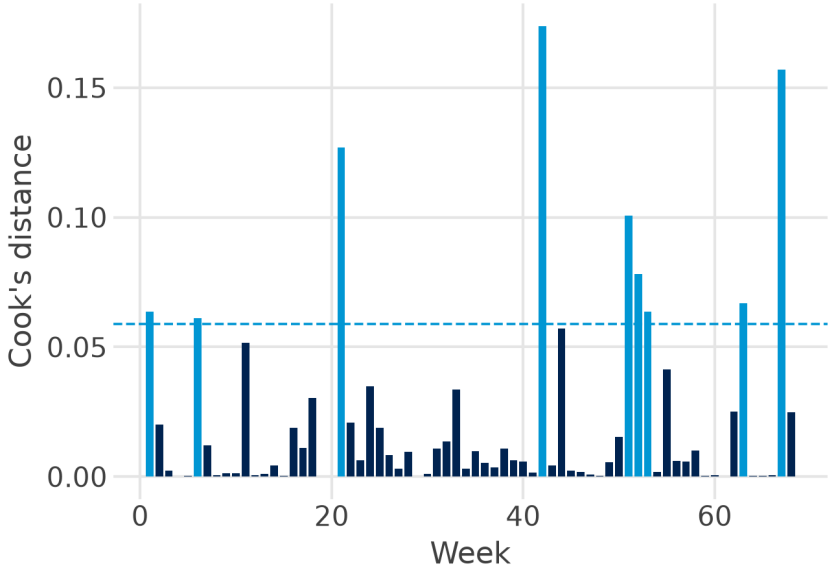
Above 5: warning; above 10: problem

Effect: large standard errors, unstable signs

Log-log model	VIF
ln price1	2.3
feature1	2.4
display1	1.1
… plus fand1:	
display1	7.6
fand1	7.8

Only three display weeks, mostly together with features: their effects cannot be separated.

6 Influential weeks: Cook's distance



Flag: 4/n = 0.059; nine weeks above

Week 42: 0.17; display 70 %, feature 42 %

Without it: elasticity −3.76 (−3.68)

But: display effect 0.45 instead of 0.61

Never delete silently: check the data, report with and without the week.

The scorecard: logs fix three assumptions at once

Test	H0	Linear	Log-log
RESET p	right form	< 0.001 ✗	0.32 ✓
Breusch-Pagan p	constant variance	< 0.001 ✗	0.68 ✓
Jarque-Bera p	normal residuals	< 0.001 ✗	0.50 ✓
Shapiro-Wilk p	normal residuals	0.001 ✗	0.50 ✓
Largest VIF	below 5	2.3 ✓	2.4 ✓
Durbin-Watson	about 2	1.48 ✗	1.13 ✗

Report the log-log model: sales respond in per cent, so errors grow with the level; logs fix that.

Dummy coding

Part 2 of 6 · 0/1 variables, categories, the reference group, the dummy trap

A dummy: 1 if yes, 0 if no

Week	december	sales
1	1	66
2	0	86
…	…	…
53	1	70

Linear: December −25 units, p = 0.32

Log-log: b = −0.319, p = 0.02

In per cent: e^b − 1 = e^−0.319 − 1 = −27 %

Reading a dummy: the difference to the weeks with 0, other variables held constant.

Small b (|b| < 0.1): 100 × b is close enough; larger b: use e^b − 1

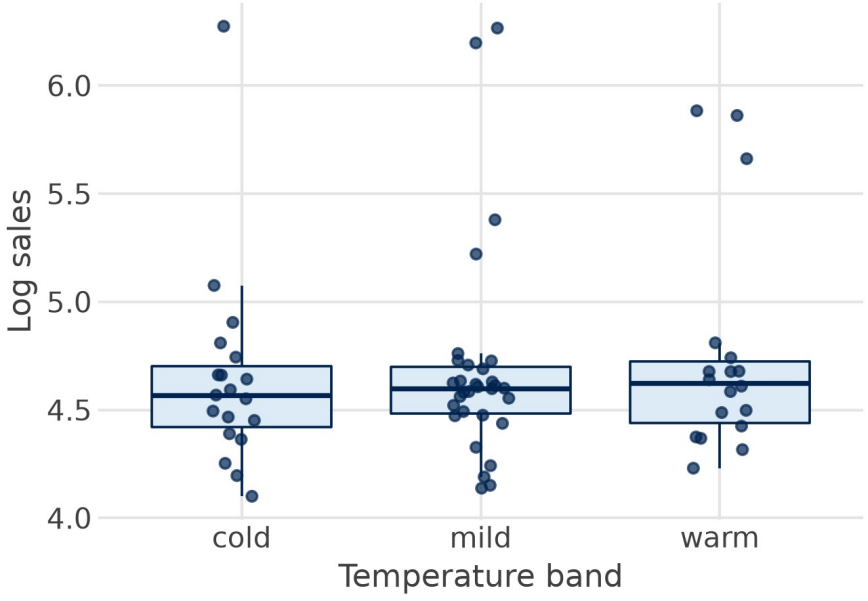
A category with k groups needs k − 1 dummies

Band	Weeks	cold	warm
cold, < 5 °C	19	1	0
mild, 5 to 14 °C	31	0	0
warm, ≥ 15 °C	18	0	1

mild = reference: the row of zeros

cold vs mild: +0.4 %, p = 0.94

warm vs mild: −7.1 %, p = 0.18



Result: with price and December in the model, the season adds little (F-test p = 0.33).

The dummy trap and the choice of reference

The dummy trap

cold + mild + warm = 1 in every week

= the intercept column: perfect collinearity

Always leave one group out

The reference group

Coefficients: differences to it

Changes the reading, not the fit (R^2 0.900)

Pick the normal case: mild

statsmodels formula

```
C(season)                          # reference: cold, first in the alphabet
C(season, Treatment("mild"))       # reference: mild
```

statsmodels avoids the trap when you write C(); with hand-made dummies, leave one out yourself.

Three pitfalls with dummies

One Easter week

- 1 in one week only
- fits it exactly (leverage 1.00)
- no real Easter effect

3 display weeks

- b = 0.61 rests on 3 weeks
- without week 42: 0.45
- count the 1s first

Level, not slope

- a dummy shifts the line
- a different slope needs
- an interaction

Before you trust a dummy: count how many 1s it has, e.g. (choc["display1"] > 0).sum().

Log-log regression

Part 3 of 6 · why logs, four log forms, reading and comparing them

Why logs: sales respond in per cent

```
sales    = a · price1^b · e^(c·december) · error          (multiplicative)
ln sales = ln a + b · ln price1 + c · december + ln error  (linear in logs)
```

Elasticities

b: per cent change in sales for 1 % more price

Comparable

across brands, markets and countries

Better errors

constant in per cent: the funnel disappears

No log of zero: shares with zeros (feature1 in 22 weeks) and dummies stay in levels.

Sources: Skiera, Reiner & Albers 2022; Hanssens, Parsons & Schultz 2001

Four forms, four readings of b

Form	Equation	b means	Brand 1
Linear	sales = a + b·price	+1 EUR: b units	−669: 10 cents, −67 units
Log-log	ln sales = a + b·ln price	+1 %: b % (elasticity)	−3.94: 1 %, −3.9 %
Log-linear	ln sales = a + b·price	+1: (e^b − 1)·100 %	−3.10: 10 cents, −27 %
Linear-log	sales = a + b·ln price	+1 %: b/100 units	−853: 1 %, −8.5 units

Log on the left: per cent change in sales. Log on the right: per cent change in the driver.

Sources: Wooldridge 2025, ch. 6; Skiera, Reiner & Albers 2022

Comparing a linear and a log model

Not R^2 or AIC: sales and log sales are different outcomes (0.884 vs 0.835 says nothing)

Residuals and tests: curve, funnel, Q-Q plot, RESET, Breusch-Pagan (Part 1)

Plausibility: sign and size of the coefficients, benchmarks such as the mean elasticity −2.6

Back on the sales scale: exp(fitted) for the log model, then compare predictions in units

Brand 1: the log-log model passes the tests, the linear one fails three. Report the log-log model.

Sources: Wooldridge 2025, ch. 6; Bijmolt, van Heerde & Pieters 2005

Non-linear relationships

Part 4 of 6 · curves in a linear regression:
squares, logs, the turning point

Linear in b, not in x

Diminishing

each step adds less

logs; square root

U or inverted U

an optimum or minimum

x and x^2

Thresholds

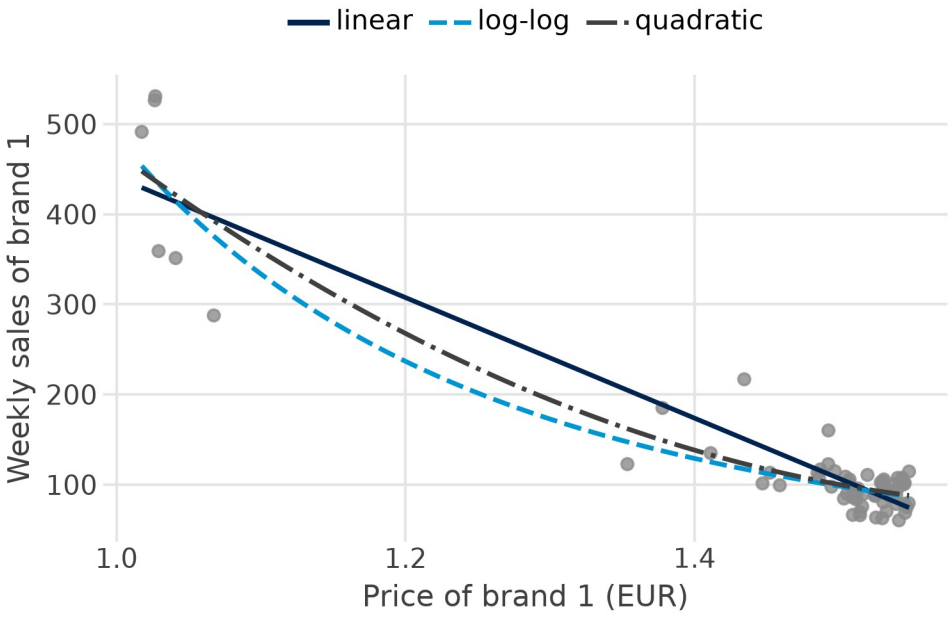
effect only above a level

dummies for ranges

```
sales = b0 + b1·price1 + b2·price1²        still linear in b0, b1, b2
smf.ols("sales ~ price1 + I(price1**2)", data=df)
```

Session 2: the marketing mix model adds saturation curves for media: the same idea.

Three curves through the same weeks



Linear: R^2 0.884; RESET p = 0.002

Quadratic: R^2 0.900; RESET p < 0.001

Log-log: R^2 0.835 (log sales); RESET p = 0.37

Thin middle: few weeks between 1.10 and 1.35

Look: all three agree where the data are; they differ where weeks are few.

The quadratic: the effect depends on the price

$$\text{sales} = 2{,}533 - 2{,}952 \cdot \text{price1} + 886 \cdot \text{price1}^2 \qquad \text{slope} = b1 + 2 \cdot b2 \cdot \text{price1}$$

Price (EUR)	Slope per EUR	10 cents more
1.00	−1,179	−118 units
1.25	−736	−74 units
1.50	−293	−29 units

1.67 turning point −b1 / (2·b2), in EUR: outside the data (1.02 to 1.55)

⚠ **Never read a curve beyond the data:** above 1.67 the quadratic says sales rise with price.

Still: RESET and Breusch-Pagan fail; the log-log model remains the better form

Choosing the form: four questions

1. **Theory:** saturation, an optimum, effects in per cent?
2. **Look and test:** residuals against fitted, RESET
3. **Compare like with like:** same outcome: adjusted R^2 or AIC; else tests and predictions
4. **Plausible in the data range:** signs, turning points, no extrapolation

Example: temp² in the full log-log model, p = 0.14: keep temperature linear.

The analyses in Python

Part 5 of 6 · packages, two models, plots, tests, dummies, curves

The packages and functions

Task	Package	Function
Fit the models	statsmodels	smf.ols(formula, data).fit()
Curve: RESET	statsmodels	linear_reset(model, power=2, use_f=True)
Funnel: Breusch-Pagan	statsmodels	het_breuschpagan(resid, exog)
Normality	statsmodels, scipy	jarque_bera(resid); stats.shapiro(resid)
Multicollinearity	statsmodels	variance_inflation_factor(exog, i)
Influence	statsmodels	model.get_influence().cooks_distance
Dummies	polars, statsmodels	pl.when().then(); C(season, Treatment("mild"))
Plots	plotnine	geom_point, geom_qq, geom_qq_line

Step 1: fit both models

Python: polars and statsmodels

```
import numpy as np
import polars as pl
import statsmodels.formula.api as smf

choc = pl.read_csv(
    "data/legacy/chocolate_dataset.csv")
df = choc.to_pandas()

lin = smf.ols("sales ~ price1 + feature1 + display1"
              " + temp + december", data=df).fit()
ll = smf.ols("np.log(sales) ~ np.log(price1)"
             " + np.log(price2) + np.log(price3)"
             " + np.log(price4) + feature1 + display1"
             " + temp + december", data=df).fit()
```

	Linear	Log-log
Own price	−678	−3.68
R^2	0.896	0.903
n	68	68

Keep both: every next step uses lin and ll.

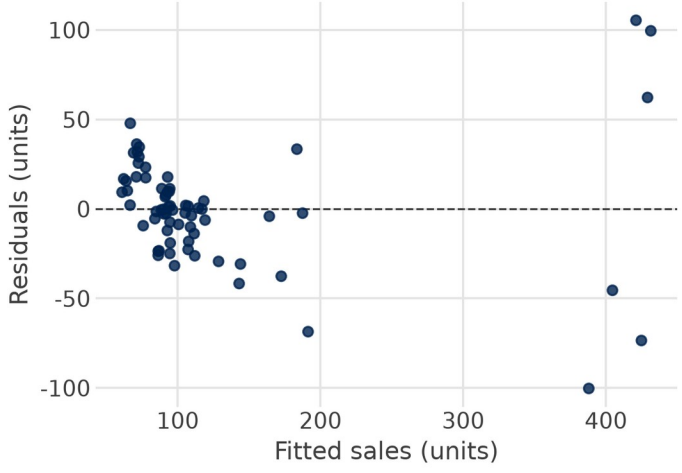
Step 2: look at the residuals

Python: plotnine

```
from plotnine import (aes, geom_hline, geom_point,
    geom_qq, geom_qq_line, ggplot, labs)

df["fit"], df["res"] = lin.fittedvalues, lin.resid
(ggplot(df, aes("fit", "res"))
 + geom_point()
 + geom_hline(yintercept=0, linetype="dashed")
 + labs(x="Fitted sales", y="Residuals"))

df["z"] = (df["res"] - df["res"].mean()) / df["res"].std()
(ggplot(df, aes(sample="z"))
 + geom_qq() + geom_qq_line()
 + labs(x="Normal quantiles", y="Residuals"))
```



lin: curve and funnel. Then the same for ll.

Run it twice: for lin and for ll; compare curve, funnel and the Q-Q line.

Step 3: test the assumptions

Python: statsmodels and scipy

```
from scipy import stats
from statsmodels.stats.diagnostic import (
    het_breuschpagan, linear_reset)
from statsmodels.stats.stattools import jarque_bera
from statsmodels.stats.outliers_influence import (
    variance_inflation_factor)

for m in (lin, ll):
    X = m.model.exog
    print(linear_reset(m, power=2, use_f=True).pvalue,
          het_breuschpagan(m.resid, X)[1],
          jarque_bera(m.resid)[1],
          stats.shapiro(m.resid).pvalue,
          max(variance_inflation_factor(X, i)
              for i in range(1, X.shape[1])))
```

p / VIF	lin	ll
RESET	< .001	0.32
BP	< .001	0.68
JB	< .001	0.50
SW	0.001	0.50
max VIF	2.3	2.4



Robust errors if needed: lin.model.fit(cov_type="HC3").

Step 4: dummies from a category

Python: polars and statsmodels

```
choc = choc.with_columns(
    pl.when(pl.col("temp") < 5).then(pl.lit("cold"))
    .when(pl.col("temp") < 15).then(pl.lit("mild"))
    .otherwise(pl.lit("warm")).alias("season"))
choc["season"].value_counts()      # 19, 31, 18
df = choc.to_pandas()

sea = smf.ols("np.log(sales) ~ np.log(price1)"
              " + np.log(price2) + np.log(price3)"
              " + np.log(price4) + feature1 + display1"
              " + december"
              " + C(season, Treatment('mild'))",
              data=df).fit()
np.exp(sea.params) - 1             # per cent
```

vs mild	e^b − 1	p
cold	+0.4 %	0.94
warm	−7.1 %	0.18
december	−28.5 %	0.01

k − 1 dummies: C() leaves the reference out for you.

Step 5: a curve and its turning point

Python: statsmodels

```
quad = smf.ols("sales ~ price1 + I(price1**2)",
               data=df).fit()
b1 = quad.params["price1"]
b2 = quad.params["I(price1 ** 2)"]
turn = -b1 / (2 * b2)                # 1.67
choc["price1"].min(), choc["price1"].max()

for p in (1.00, 1.25, 1.50):         # slope at p
    print(p, (b1 + 2 * b2 * p) / 10)  # per 10 cents

linear_reset(quad, power=2, use_f=True).pvalue
```

Result	Value
b1	−2,952
b2	886
Turning point	1.67 EUR
Price range	1.02 to 1.55
RESET p	< 0.001

Outside the data: the turning point says nothing about real prices.

In-class exercise

Part 6 of 6 · online quiz, then the chocolate data in Python, then with Posit Assistant

Online quiz 3: assumptions, dummies, logs in five questions

1 Breusch-Pagan gives p = 0.001. This means …
A constant variance B unequal variance (heteroscedasticity) C normal residuals

2 Four store formats in one model with an intercept need …
A 4 dummies B 3 dummies C 1 dummy

3 In a log model the December dummy is b = −0.36. December weeks sell …
A 0.36 units fewer B 36 % fewer C about 30 % fewer

4 A quadratic in price turns at 1.67 EUR; prices in the data run 1.02 to 1.55. So …
A do not read it: outside the data B raise the price above 1.67 C sales rise with price

5 Linear model R^2 0.88, log-log R^2 0.84. Which is better?
A the linear one B R^2 cannot tell: different outcomes C the log-log one

Online on Canvas: 3 minutes alone, then compare with your neighbour.

Your turn: which model may brand 1 trust?

Pairs, with packages: 15 minutes	Plenary, answers: 5 minutes	Posit Assistant: 10 minutes

1. **Look:** sales ~ price1 and np.log(sales) ~ np.log(price1); residual plot of each: curve, funnel?
2. **Test:** RESET, Breusch-Pagan and Jarque-Bera for both models; which fails what?
3. **Dummies:** season from temp with polars; add december and C(season, Treatment("mild")); read both in per cent
4. **Curve:** sales ~ price1 + I(price1**2); turning point; inside the price range?
5. **The manager:** which model do you report, and one sentence with a number

File: `sessions/01-foundations/assumptions_exercise.qmd` (data, tasks, Posit Assistant)

Then: the same tasks with Posit Assistant

1. **Code first:** solve the tasks with the packages; keep the cells run, so Posit Assistant sees choc, df and the models
2. **Open the chat:** Posit Assistant icon in the activity bar, or Ctrl/Cmd + Shift + P → View: Show Posit Assistant
3. **Give context:** type @assumptions_exercise.qmd, then one task per prompt: the file, the packages, the output
4. **Ask for meaning:** let it explain each test result in one sentence; check the direction of p
5. **Compare:** its code, numbers and wording with yours and the answers

Stay in Normal mode: never /yolo. Read every edit before you accept it. You own every number and every sentence.

Prompts for the assumption checks

Prompt for task 2

```
@assumptions_exercise.qmd Task 2.
Run RESET, Breusch-Pagan and
Jarque-Bera with statsmodels for
lin1 and ll1. Show the p-values
in one table and explain in one
sentence each which assumption
fails.
```

Prompt for task 3

```
Task 3. Build season from temp
with polars when/then (cold < 5,
mild < 15, warm). Fit the model
with C(season, Treatment("mild"))
and give each dummy in per cent
with exp(b) - 1.
```

Name three things: the file, the packages (polars, statsmodels, scipy), the output

Check the direction: in an assumption test a small p means a problem

Watch for slips: tests on sales instead of residuals, three dummies plus intercept, 100 × b

Key takeaways

1. **Look, then test, then fix:** residual and Q-Q plots; RESET, Breusch-Pagan, Jarque-Bera, VIF, Cook's distance
2. **In assumption tests** a small p is bad news
3. **Dummies:** k − 1 for k groups; each b is a difference to the reference; count the 1s
4. **Logs:** elasticities, per cent effects, better residuals; compare models by tests, not R^2
5. **Curves:** squares and logs keep OLS linear in b; never read a curve beyond the data

Homework: coding exercise 2 with your personal data, brief on Canvas.

References

Academic

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Bojinov, I. I., Parzen, M. & Hamilton, P. J. (2025). Linear Regression. HBS note 9-622-100 (Canvas).

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Hanssens, D. M., Parsons, L. J. & Schultz, R. L. (2001). Market Response Models. Kluwer.

Hair, J. F. et al. (2019). Multivariate Data Analysis, 8th ed. Cengage. Ch. 5.

Learning resources

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statsmodels. Regression diagnostics and specification tests (documentation).

Gelman, Hill & Vehtari (2020). Regression and Other Stories, ch. 11 to 12 (free PDF).

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